

Beyond First-Order Drift: Practical WSPR-2 Decoding Robustness under Nonstationary Frequency Instability

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ABSTRACT Frequency error in a weak-signal transmission has components that a receiver may search explicitly and components outside its trajectory model. We measured this distinction for a generated WSPR-2 signal decoded by the unmodified production WSJT-X 3.0.2 `wspird`. The independent waveform implementation matched all 162 reference channel symbols; complex additive white Gaussian noise (AWGN) was calibrated to the 2500-Hz WSPR SNR convention. The AWGN-only 50% decode point was -31.519 dB by interpolation. Exact decoder-aligned linear drift, an orthogonal quadratic residual, thermal-settling shape probes, and stochastic frequency-modulation (FM) families were evaluated on saved, seeded trial grids. Signed linear-drift 50% boundaries contracted from $+0.698/-0.711$ Hz at -30.0 dB to $+0.299/-0.331$ Hz at -31.0 dB. The corresponding orthogonal-quadratic 50% amplitude decreased from 0.441 to 0.203 Hz. A balanced factorial fit measured a linear-drift $\times$ residual interaction coefficient of 10.16 log-odds/Hz² on its fitted scale. No single tested scalar residual descriptor performed uniformly across trajectory families: maximum phase excursion improved prediction for smooth deterministic shapes, whereas residual-frequency RMS scored better on the held-out random-walk family. A further 5,400-trial sweep of synthetic Allan-targeted FM processes related fractional-frequency stability, carrier scaling, projected residuals, and decode outcomes. The results characterize this decoder and these generated waveforms and grids; no transmitter hardware, measured oscillator, or propagation impairment was tested.

INDEX TERMS WSPR-2, weak-signal decoding, frequency drift, oscillator frequency noise, Allan deviation, model mismatch.

I. INTRODUCTION

WSPR-2 combines long symbol duration with low-rate coding for reception at weak signal-to-noise ratio (SNR).¹ The original WSPR account describes its narrow four-tone signal and propagation-beacon purpose [1], and the current official guide documents its protocol and 2500-Hz SNR reference [2]. Frequency variation across the active frame can be represented by a center-frequency offset, a linear drift, or more complex time dependence. Those categories matter because the tested production decoder searches constant frequency and a linear symbol-index drift [3], while an imposed residual orthogonal to both remains outside that search trajectory class.

Related work addresses distinct observables. A non-refereed WSPR-2 transmitter spectral-width article provides

technical background on reference stability and frequency-generation choices [4]. In communications research, measured oscillator phase-noise spectra have been related to performance, and discrete-time phase-noise models to residual phase error after tracking [5], [6]; these are not WSPR decoder studies. Allan and later frequency-stability analyses provide the metrology basis for power-law noise classification [7]–[9]. FST4W spectral-spread reports have been used to examine propagation modes [10], while WSPRNet and other amateur-radio reports have supported HF propagation sensing [11]–[13].

These studies motivate attention to frequency stability and time-varying spectral behavior. The present work measures decode-probability boundaries for production `wspird` under controlled in-model and beyond-model frequency trajectories and compares residual-severity descriptors over the tested families. It characterizes an implementation-specific observation envelope, not a decoder-independent WSPR limit.

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II. WSPR-2 DECODER AND EXPERIMENTAL FRAMEWORK

A. WSPR-2 WAVEFORM AND PRODUCTION DECODER

All decoder outcomes use the unmodified production `wsprd` from WSJT-X 3.0.2, with release package and matching source provenance recorded in the repository [3]. The generated WSPR-2 active frame contains 162 symbols, 256 complex samples per symbol at 375 samples/s, and therefore lasts 110.592 s. The C2 input contains the required timing padding. The decoder source searches center frequency and a symbol-index linear drift; this is a property of this implementation, not an assertion about all WSPR receivers. The decoded Type-1 message is K1ABC FN42 33.

B. INDEPENDENT WAVEFORM VALIDATION

The Python encoder's 162 channel symbols matched the official WSJT-X `wsprsim` reference vector at all 162 positions. Its generated clean C2 signal was decoded by production `wsprd` as K1ABC FN42 33. The source archive identity and clean-decode checks are recorded with the repository's waveform validation. This gate checks experimental input compatibility; it does not itself establish noisy-channel performance.

C. AWGN CALIBRATION AND DECODE-SUCCESS CRITERION

The repository defines injected SNR in a 2500-Hz reference bandwidth [2]. For active complex-signal power P_s , complex noise power at the 375-Hz C2 sample rate is

$$P_n = P_s \frac{375}{2500} 10^{-\text{SNR}_{2500, \text{dB}}/10}, \quad (1)$$

$$\text{Var}(n_I) = \text{Var}(n_Q) = P_n/2.$$

Independent Gaussian in-phase and quadrature components were applied with deterministic seeds. The 375/2500 factor corresponds to 8.239 dB between bandwidth conventions, compared with the official simulator's rounded 8.2-dB adjustment. For N trials, P_{decode} is the number of exact decodes of K1ABC FN42 33 divided by N ; a partial or different message is not a success. Binomial cell intervals in saved summaries are Wilson 95% intervals. Decoder-reported SNR is retained as a diagnostic rather than substituted for injected SNR.

D. FREQUENCY-TRAJECTORY DECOMPOSITION

At symbol index $i = 0, \dots, 161$, define $x_i = (i - 81)/81$ and the design matrix $X = [1, x]$. A symbol-rate frequency trajectory is decomposed as

$$\Delta f_i = c_0 + c_1 x_i + r_i, \quad (2)$$

$$\begin{bmatrix} c_0 \\ c_1 \end{bmatrix} = \arg \min_b \|\Delta f - Xb\|_2^2, \quad X^T r \approx 0.$$

The fitted constant-plus-linear component is decoder-representable in form; the projected residual is a controlled beyond-linear component. Symbol-wise frequency offsets are held for the 256 samples of each symbol and applied by

TABLE 1. AWGN Baseline Thresholds and SNR Reporting

Quantity	Value
SNR at 90% decoding	−30.286 dB
SNR at 50% decoding	−31.519 dB
SNR at 10% decoding	−32.729 dB
Mean reported minus injected SNR	+0.111 dB
Median reported minus injected SNR	0.0 dB

Thresholds are piecewise interpolations. The SNR differences use 1,201 successful decodes with reported SNR.

continuous phase accumulation. Phase descriptors use the same 375-Hz sample timing and the residual rather than the unprojected raw process. Orthogonality refers to the discrete 162-symbol $[1, x]$ basis within floating-point precision.

E. STATISTICAL METHODOLOGY

Saved experiments use stated condition grids and deterministic seeds. Where comparisons were designed with matched noise, the same seed set was reused across trajectory conditions at a fixed SNR. Trial-level outcomes, summary arithmetic, row counts, and seed-set checks are documented with their outputs. SNR50, signed D50, and A50 values are piecewise-linear interpolations between observed probability points, not extrapolations. The balanced interaction analysis fits additive (M0) and drift $\times$ residual (M1) logistic models and reports ordinary and cluster-robust uncertainty plus cluster bootstrap. Descriptor models are evaluated on held-out trajectory families and later external sets; Brier and log loss are predictive scores for the specified splits.

III. ROBUSTNESS WITHIN THE DECODER MODEL CLASS

A. EXACT FIRST-ORDER DRIFT MODEL

The decoder-aligned trajectory is

$$\Delta f_i = \frac{D}{2} x_i = \frac{D}{2} \frac{i - 81}{81}, \quad i = 0, \dots, 161. \quad (3)$$

It matches the source-inspected `wsprd` drift convention [3]. The endpoint at $i = 161$ is slightly below $+D/2$; the implementation uses the exact expression rather than a sample-wise endpoint-to-endpoint substitute. Positive and negative drift conditions remain separate in the raw results.

B. NOISELESS AND HIGH-SNR SANITY CONTROLS

Strict noiseless controls at $D = 0, \pm 1, \pm 2, \pm 3, \pm 4$ Hz decoded the expected message in each $N = 1$ condition. At injected −20 dB, all nine corresponding conditions decoded 20/20 trials. These are implementation sanity checks of the waveform and exact drift path, not precision decode-probability estimates. The same strict-control records show 141/200 at zero drift, 8/200 at +2 Hz, and 7/200 at −2 Hz at −31 dB. Robustness boundaries below come from the 100/200-trial maps, not the $N = 1$ or $N = 20$ controls.

C. SNR-DEPENDENT LINEAR-DRIFT BOUNDARY

The AWGN-only baseline had interpolated SNR90, SNR50, and SNR10 values of −30.286, −31.519, and −32.729 dB,

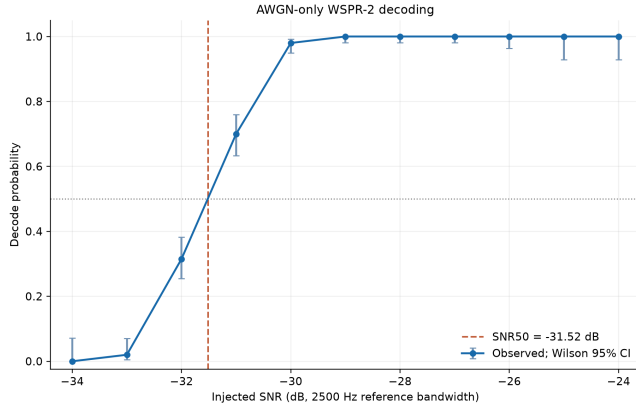


FIGURE 1. AWGN-only production- w_{sprd} P_{decode} versus injected SNR (dB, 2500-Hz reference), with Wilson intervals and an interpolated 50% point.

TABLE 2. Grid-Interpolated 50% Decode Boundaries

SNR (dB)	D_{50}^+ (Hz)	D_{50}^- (Hz)	A_{50} (Hz)
-29.5	n.b.	n.b.	0.578
-30.0	+0.698	-0.711	0.441
-30.5	+0.483	-0.489	0.311
-31.0	+0.299	-0.331	0.203
-31.5	0.0	0.0	n.t.

n.b.: not bracketed by observed drift points. n.t.: not tested. Zero at -31.5 dB is a zero-drift boundary, not a nonzero tolerance.

respectively (Table 1, Fig. 1). Across 1,201 successful decodes with reported SNR, the mean reported-minus-injected difference was +0.111 dB and the median was 0.0 dB.

The refined signed-drift map used 200 trials per SNR $\times$ drift cell. Interpolated positive/negative D50 values were +0.698/ - 0.711 Hz at -30.0 dB, +0.483/ - 0.489 Hz at -30.5 dB, and +0.299/ - 0.331 Hz at -31.0 dB (Table 2, Fig. 2). At -29.5 dB the tested drift values did not bracket 50%, so no D50 is assigned. At -31.5 dB the reported zero-drift D50 is a boundary observation where the zero-drift cell is at 50%; it is not a measured nonzero drift allowance. The source heatmap retains separate positive and negative panels.

IV. BEYOND FIRST-ORDER FREQUENCY DYNAMICS

A. ORTHOGONAL QUADRATIC RESIDUAL CONSTRUCTION

To test model mismatch without a hidden constant or linear term, set $q = x^2$, compute $\beta = \arg \min_b \|q - Xb\|_2^2$, and form

$$\begin{aligned} q_{\perp} &= q - X\beta, \\ q_{\text{norm}} &= \frac{q_{\perp}}{\max_i |q_{\perp,i}|}, \\ r_i &= A_{\text{res}} q_{\text{norm},i}. \end{aligned} \quad (4)$$

No coefficient is hard-coded. At $A_{\text{res}} = 1$ Hz, saved discrete dot products of r with the constant and x columns were -1.710×10^{-14} and 4.441×10^{-16} Hz, respectively. This is an orthogonal test shape, not a physical oscillator model. The pilot held nominal drift at zero.

TABLE 3. Balanced Linear $\times$ Quadratic Interaction

Interaction estimate (log-odds/Hz ²)	10.16
Ordinary SE	0.91
Cluster-robust SE	0.84
Cluster-bootstrap 95% interval	[8.54, 11.95]
Metric	M0 / M1
Log-likelihood	-1763.2 / -1697.7
AIC	3534.4 / 3405.3
BIC	3559.2 / 3436.3
Brier score	0.1607 / 0.1538

18 cells, 200 trials/cell; matched noise seeds within SNR. M0/M1 scores are in sample.

B. DECODE-PROBABILITY BOUNDARY

The quadratic pilot used 100 trials per tested SNR $\times$ amplitude cell. Its bracketed A50 estimates were 0.578, 0.441, 0.311, and 0.203 Hz at -29.5, -30.0, -30.5, and -31.0 dB, respectively (Table 2, Fig. 3). They measure the amplitude of this normalized basis at the 50% crossing, not a threshold for arbitrary shapes.

C. LINEAR $\times$ QUADRATIC NON-ADDITIVITY

The balanced factorial set tested SNR -30.0/ -30.5/ -31.0 dB, $D = 0/0.5$ Hz, and $A_{\text{res}} = 0/0.25/0.5$ Hz, with 200 trials in each of 18 cells (3,600 trials). Within each SNR the six conditions shared matched noise seeds. M1 added $D \times A_{\text{res}}$ to the SNR, D , and A main effects. Its fitted interaction coefficient was 10.16 log-odds/Hz² (ordinary SE 0.91; cluster-robust SE 0.84; 1,000-fit cluster-bootstrap 95% interval [8.54, 11.95]). M0 versus M1 AIC was 3534.4 versus 3405.3, BIC 3559.2 versus 3436.3, and in-sample Brier 0.1607 versus 0.1538 (Table 3, Fig. 4). This is non-additivity on the fitted log-odds scale. The coefficient is specific to the tested factorial grid, parameter scaling, and logit specification; it is not a transferable physical constant, and the fit alone does not identify a decoder mechanism.

V. PHYSICALLY MOTIVATED FREQUENCY-INSTABILITY MODELS

A. THERMAL-SETTLING RESIDUALS

A deterministic shape probe evaluates

$$f_{\text{raw}}(t) = K [1 - \exp(-t/\tau)] \quad (5)$$

at the 162 symbol-center times and removes its least-squares $[1, x]$ component. For $K = 1$ Hz, residual peak amplitudes $g_{\tau} = \max |r|$ are 0.323, 0.141, and 0.049 Hz for $\tau = 30, 60, 120$ s. Their normalized correlations with the *negative* quadratic residual are 0.957, 0.988, and 0.997. The time constants are shape probes. Amplitude-matched thermal and signed-quadratic trials used matched noise seeds. At -30 dB and a 0.5-Hz peak residual, observed thermal decode probabilities were 0.68, 0.50, and 0.46 for $\tau = 30, 60, 120$ s; signed-quadratic controls were 0.35 for positive and 0.36 for negative amplitude. Shape correlation alone does not specify equal measured decode probability in these cells (Fig. 5).

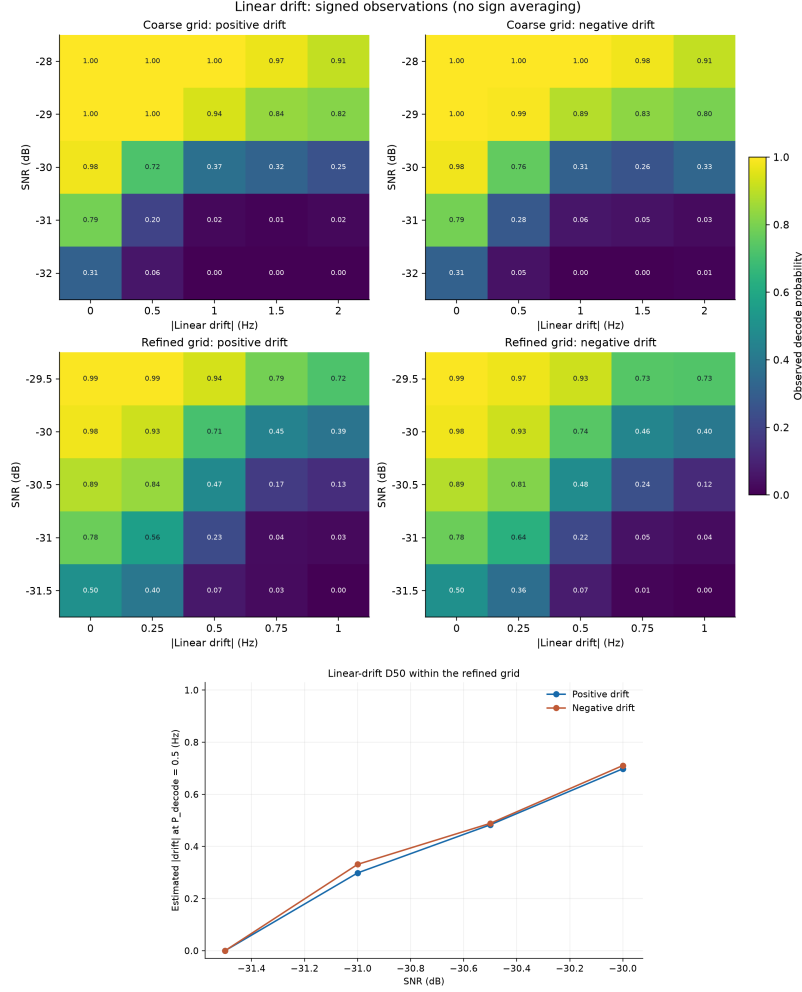


FIGURE 2. Linear-drift response. Top: coarse/refined P_{decode} maps with positive and negative drift in separate panels and no sign averaging. Bottom: signed, grid-interpolated D50 (Hz) versus injected SNR (dB).

B. STOCHASTIC RANDOM-WALK FM

A separate, uncalibrated symbol-rate probe starts at $f_{\text{raw},0} = 0$ and takes Gaussian increments,

$$f_{\text{raw},i} = f_{\text{raw},i-1} + \epsilon_i, \quad \epsilon_i \sim \mathcal{N}(0, \sigma_{\text{step}}^2), \quad (6)$$

before $[1, x]$ projection. Preflight medians of projected RMS frequency were 0.0151, 0.0299, 0.0600, 0.1207, and 0.2389 Hz for $\sigma_{\text{step}} = 0.005, 0.01, 0.02, 0.04, 0.08$ Hz. Decoder tests used the latter four levels, 200 trials per cell. At SNR -30.0 dB, observed probabilities were 0.935/0.945/0.815/0.345 in ascending σ_{step} ; at -30.5 dB, 0.885/0.795/0.675/0.175; and at -31.0 dB, 0.745/0.620/0.355/0.035. This random walk is an uncalibrated stochastic trajectory probe.

C. ALLAN-DEVIATION-PARAMETERIZED WHITE/FLICKER/RANDOM-WALK FM

The later synthetic FM preflight generated 1-s-cadence fractional-frequency processes $y(t)$ of white, flicker, and random-walk type, scaled to ensemble targets $\sigma_y(1s) =$

$10^{-10}, 3 \times 10^{-10}, 10^{-9}, 3 \times 10^{-9}, 10^{-8}$ over carriers 7/10/14/28 MHz. It used 500 independent trajectories per family $\times$ carrier $\times$ stability cell. The process was sampled by zero-order hold at validated WSPR symbol centers, mapped through

$$\Delta f(t) = f_c y(t), \quad (7)$$

and projected off $[1, x]$. The preflight's fitted raw Allan-deviation slopes were $-0.503, -0.028,$ and $+0.439$ for white, flicker, and random-walk FM; corresponding projected slopes were $-0.434, +0.012,$ and $+0.437$. They satisfy the configured ± 0.18 qualitative-slope check against $-1/2, 0,$ and $+1/2$, respectively [8], [9] (Fig. 6). These are finite-cadence synthetic generators, not traceable metrology references.

D. CARRIER-FREQUENCY SCALING

The production-decoder physical-parameter sweep selected carriers 10/14/28 MHz, targets $\sigma_y(1s) = 10^{-9}, 3 \times 10^{-9}, 10^{-8}$, and SNR $-30.5/-31.0$ dB, with 100 trials in each of 54 cells (5,400 trials). The oscillator trajectory was reused across the two SNR values and AWGN seeds matched

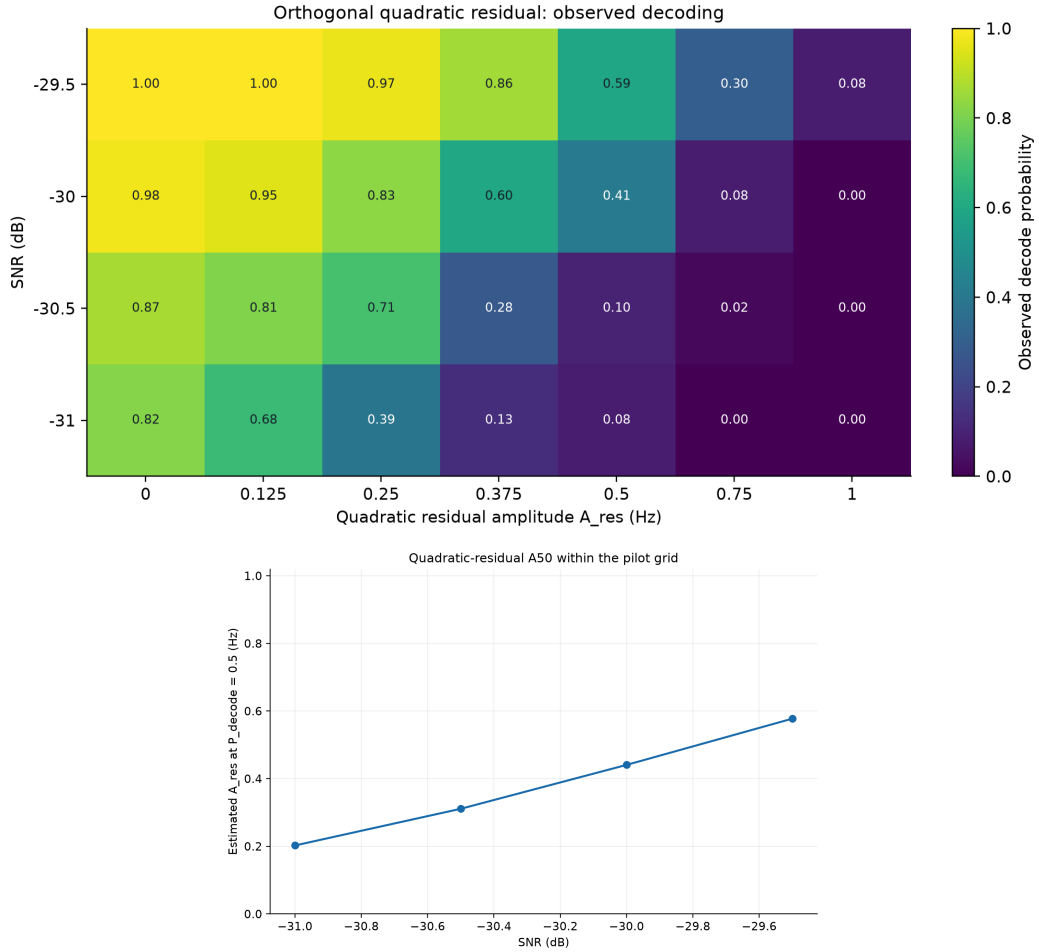


FIGURE 3. Orthogonal-quadratic response. Top: observed P_{decode} by residual amplitude (Hz) and injected SNR (dB). Bottom: grid-interpolated A50 (Hz) where the 50% crossing is bracketed.

across physical cells within an SNR. At 10 MHz, target 10^{-8} , and -31 dB, median projected frequency RMS was 0.097/0.156/0.345 Hz for white/flicker/random-walk FM, with observed $P_{\text{decode}} = 0.53/0.22/0.01$. For random-walk FM at target 3×10^{-9} and -31 dB, the 10/14/28-MHz cells had median RMS 0.113/0.142/0.278 Hz and probabilities 0.43/0.27/0.02 (Table 4, Fig. 7). The carrier variable is the fractional-to-hertz scaling parameter, not a comparison of measured transmitters.

VI. CROSS-TRAJECTORY SEVERITY DESCRIPTORS

A. PEAK RESIDUAL FREQUENCY

The AMP candidate uses the largest absolute value of projected residual frequency. It is easy to compute, but equal peaks can arise from different time histories. Evaluation used identical held-out trials for the scalar candidates within each comparison (Table 5).

B. MAXIMUM RESIDUAL PHASE EXCURSION

Residual phase is the continuous accumulation of projected frequency:

$$\phi[n+1] = \phi[n] + \frac{2\pi}{375} r_f[n], \quad (8)$$

with each symbol offset held across its 256 samples. PHASE augments AMP with $\max_n |\phi[n]|$. In five-family leave-one-family-out tests over smooth deterministic quadratic-positive, quadratic-negative, and three thermal shapes (3,000 held-out trial predictions), PHASE achieved Brier 0.1677 and log loss 0.5060, compared with AMP 0.1737 and 0.5228. The paired AMP-minus-PHASE Brier difference was $+0.0060$, with a cluster-bootstrap 95% interval $[+0.0035, +0.0085]$.

C. RESIDUAL-FREQUENCY RMS

RMS uses

$$R_f = \left(\frac{1}{162} \sum_{i=0}^{161} r_i^2 \right)^{1/2} \quad (9)$$

after projection. The separately evaluated SNR-plus-RMS model scored Brier 0.1681 and log loss 0.5071 in smooth-family holdouts. None of AMP, PHASE, or RMS describes

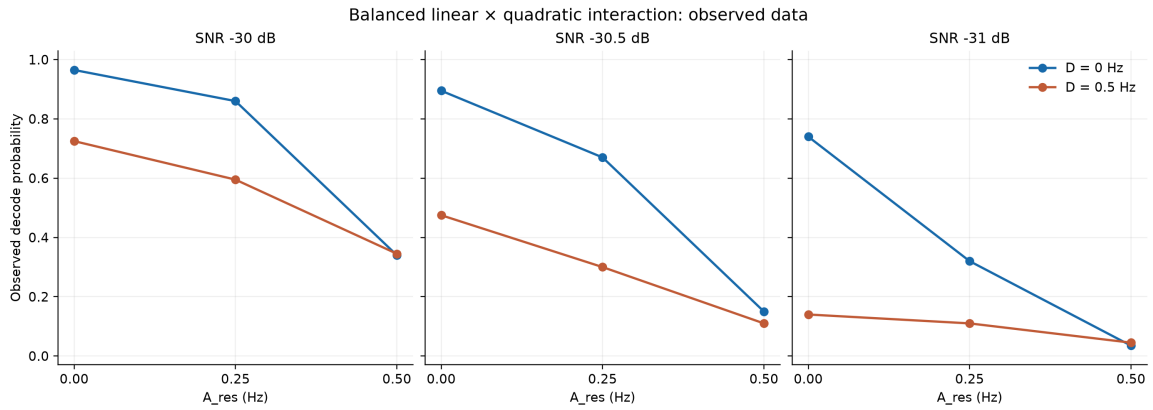


FIGURE 4. Observed balanced-factorial P_{decode} versus quadratic amplitude (Hz), separated by exact linear drift and injected SNR (dB).

TABLE 4. Representative Synthetic Oscillator Decoder Cells at Injected SNR –31.0 dB

Noise family	Carrier (MHz)	Target $\sigma_y(1\text{ s})$	Median residual RMS (Hz)	Correct/100	P_{decode}
White FM	10	1×10^{-8}	0.097	53	0.53
Flicker FM	10	1×10^{-8}	0.156	22	0.22
Random-walk FM	10	1×10^{-8}	0.345	1	0.01
Random-walk FM	10	3×10^{-9}	0.113	43	0.43
Random-walk FM	14	3×10^{-9}	0.142	27	0.27
Random-walk FM	28	3×10^{-9}	0.278	2	0.02

TABLE 5. Held-Out Descriptor-Model Scores

Evaluation	Metric	AMP	PHASE	RMS
Smooth LOFO	Brier	.1737	.1677	.1681
3,000 trials	Log loss	.5228	.5060	.5071
Random walk	Brier	.1480	.1614	.1453
2,400 trials	Log loss	.4668	.5017	.4575

AMP: peak projected frequency. PHASE: AMP plus maximum projected phase. RMS: SNR plus projected frequency RMS.

the entire temporal trajectory; differences among amplitude-matched thermal and quadratic cells illustrate this limitation.

D. HELD-OUT AND EXTERNAL VALIDATION

On the independent 2,400-trial symbol-rate random-walk set, frozen AMP/PHASE/RMS models scored Brier 0.1480/0.1614/0.1453 and log loss 0.4668/0.5017/0.4575 (Table 5). Thus PHASE's improvement over AMP in the tested smooth-shape holdouts did not persist on that random-walk set, while RMS had the lowest score among these three tested scalars there. On a further 5,400 synthetic Allan-targeted physical-sweep trials, the frozen SNR-plus-RMS model scored Brier 0.1440 and log loss 0.4415; those outcomes were not used to refit it. The external score is tied to this sweep's outcome prevalence and support.

E. REMAINING MORPHOLOGY DEPENDENCE

The six-family RMS support audit found that 1,467 of 2,400 held-out random-walk rows were outside the training families' RMS support. A diagnostic all-data fit with family terms

had AIC 5178.2 versus 5177.9 without them, BIC 5230.9 versus 5197.6, and Brier 0.1562 versus 0.1566. That fit is descriptive rather than a deployable predictor. Family-specific residual errors and out-of-support predictions remain. No single tested scalar descriptor was uniformly best: PHASE had the lowest scores among the tested candidates in smooth-family holdouts, whereas RMS had the lowest scores on the held-out random-walk set; neither removed family and support dependence.

VII. ENGINEERING IMPLICATIONS

For this production decoder and generated-signal test bed, frequency-stability assessment can be organized as fractional frequency $y(t)$, absolute deviation $f_c y(t)$, $[1, x]$ projection, residual descriptor, and observed decode probability. This distinguishes a constant or linear component the decoder searches from residual motion it does not explicitly represent. The saved carrier-scaling audit records higher projected hertz severity at higher f_c for comparable fractional targets, while the physical-sweep cells show that FM family and injected SNR also change observed probabilities. A specification expressed solely as total drift, peak offset, or Allan deviation at one averaging time would not reproduce the tested cell outcomes without process shape, carrier scaling, and decoder configuration.

VIII. LIMITATIONS

Every decode probability belongs to the unmodified production WSJT-X 3.0.2 `wsprrd`, the fixed generated message/waveform, the repository's C2 path and AWGN conven-

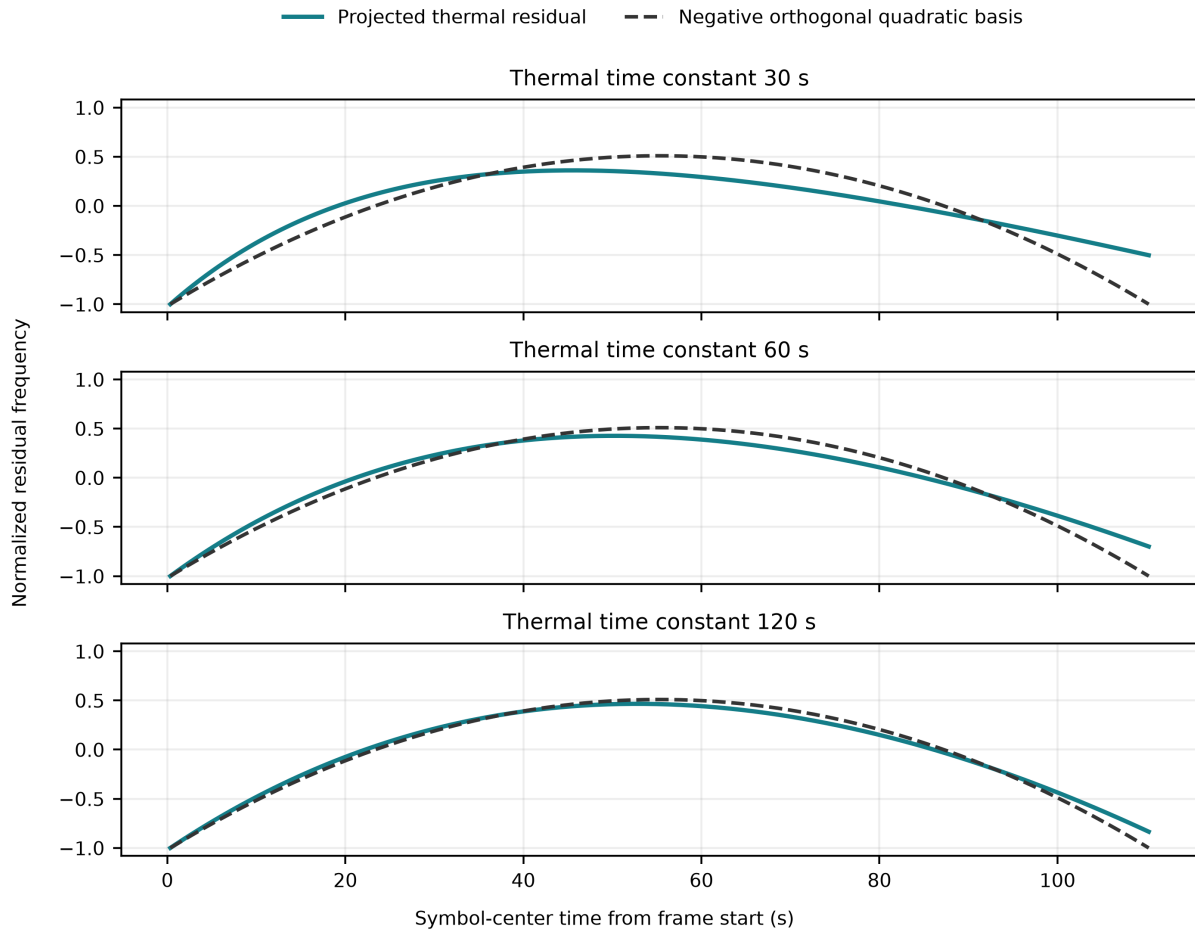


FIGURE 5. Projected and normalized thermal-settling residual shapes for the tested τ values compared with the negative controlled quadratic basis. The τ values are shape probes, not measured hardware constants.

tion, and a finite tested grid. The decoder's acquisition and candidate-search behavior is part of the outcome; an ideal receiver was not evaluated. D50 and A50 are interpolated within observed brackets and inherit Monte Carlo uncertainty. The $N = 1$ noiseless and $N = 20$ high-SNR controls are implementation checks; finite 100- or 200-trial cells are not exact probabilities.

No RF hardware, measured oscillator trace, field propagation, Doppler, or nonstationary multipath was evaluated. Thermal time constants and random-walk steps are shape probes. Allan-targeted synthetic processes have finite sample cadence and spectral support and are scaled to an ensemble 1-s statistic; qualitative slope agreement does not make them traceable frequency standards. The physical sweep's carrier labels change $\Delta f = f_{cy}$, not the C2 dial frequency. Matched seeds reduce comparison noise but do not create independent replications. Smooth-family holdouts reuse a small set of deterministic shapes, and descriptor models exhibit support and morphology dependence.

IX. CONCLUSION

For production WSJT-X 3.0.2 *wsprdx*, the tested generated signals show an SNR-dependent observation envelope both within its first-order drift model class and for projected beyond-linear residuals. The controlled quadratic grid provided bracketed A50 estimates, while the balanced factorial yielded a nonzero fitted drift $\times$ residual term on the tested log-odds scale. No single tested scalar residual descriptor performed uniformly across all trajectory families. Residual-frequency RMS was the most transferable of the tested scalars in the external stochastic comparisons, but family and support effects remained. These boundaries and scores belong to the specified decoder, waveform, and condition grids, not measured-hardware operation.

DATA AND CODE AVAILABILITY

Code, trial-level data, derived analyses, figure-generation scripts, decoder provenance, and deterministic seed specifications are available at the project repository, <https://github.com/surnamemei/wspr-beyond-linear-drift>. The staged figures in this source tree are presentation copies; the validated result

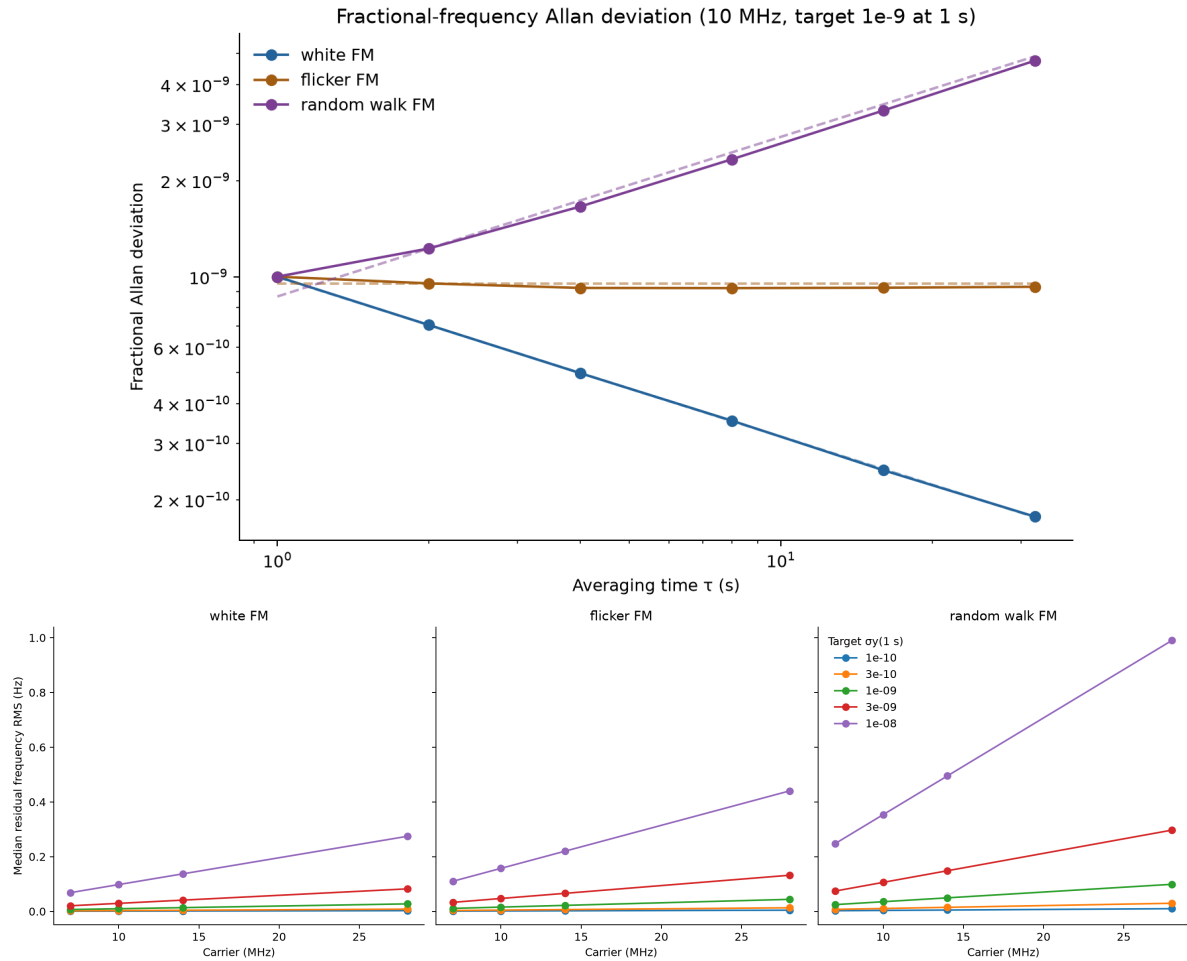


FIGURE 6. Synthetic FM preflight. Top: fractional-frequency Allan deviation versus averaging time (s), with qualitative slope guides. Bottom: projected residual-frequency RMS (Hz) versus carrier-scaling parameter (MHz).

files remain unchanged.

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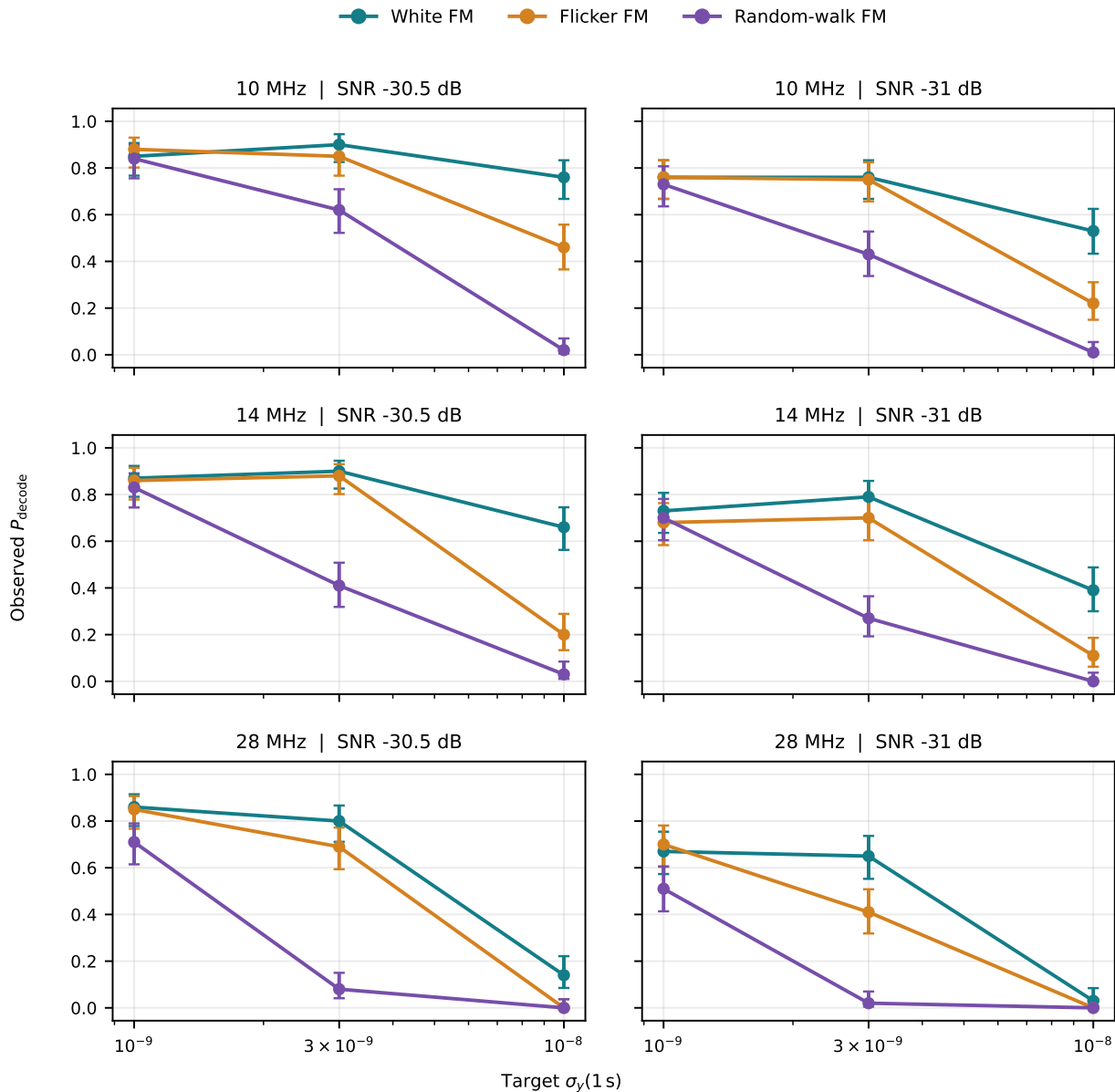


FIGURE 7. Observed production-wsprd P_{decode} in the saved 54-cell synthetic physical-parameter sweep. Rows are carrier-scaling frequencies (MHz); columns are injected SNRs (dB). Points denote 100-trial cell proportions; bars are Wilson 95% intervals.

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